Gemma3

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> **Demonstration environment**

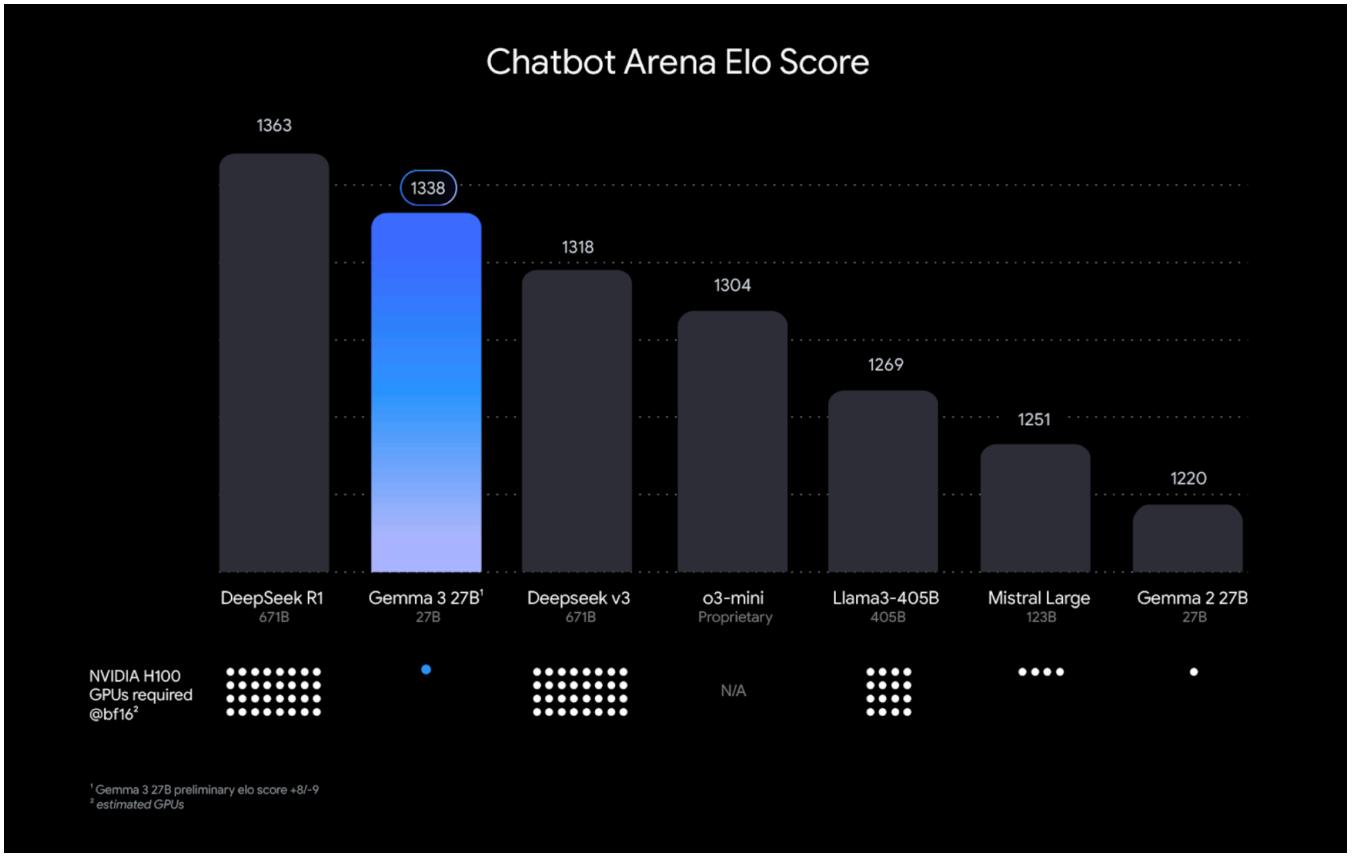
Development board: Jetson Orin series mainboard

Gemma is a family of lightweight models built by Google on top of the Gemini technology. The Gemma 3 models are multimodal (they handle both text and images), have a 128K context window and support more than 140 languages.

1. Model size

Model	Size
gemma3:1b	815MB
gemma3:4b	3.3GB

2. Performance

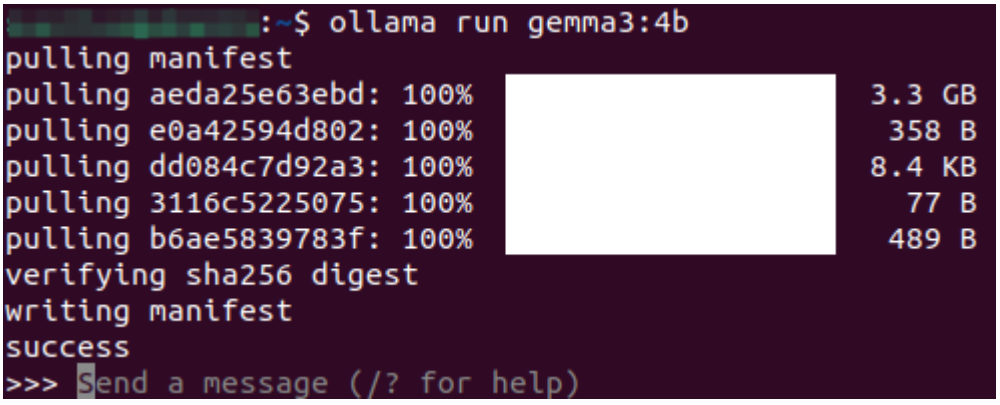


3. Using Gemma3

3.1 Run Gemma3

Use the run command to start the model; if the model has not been downloaded before, it is pulled automatically from the Ollama model library:

```
ollama run gemma3:4b
```



3.2 Start a conversation

```
How to learn a programming language?
```

The time it takes to answer depends on the hardware configuration, so be patient!

```
-------
>>> How to learn a programming language?
Learning a programming language can feel daunting at first, but with a
structured approach and consistent effort, it's absolutely achievable.
Here's a breakdown of how to learn a programming language effectively,
broken down into stages and with different learning styles in mind:

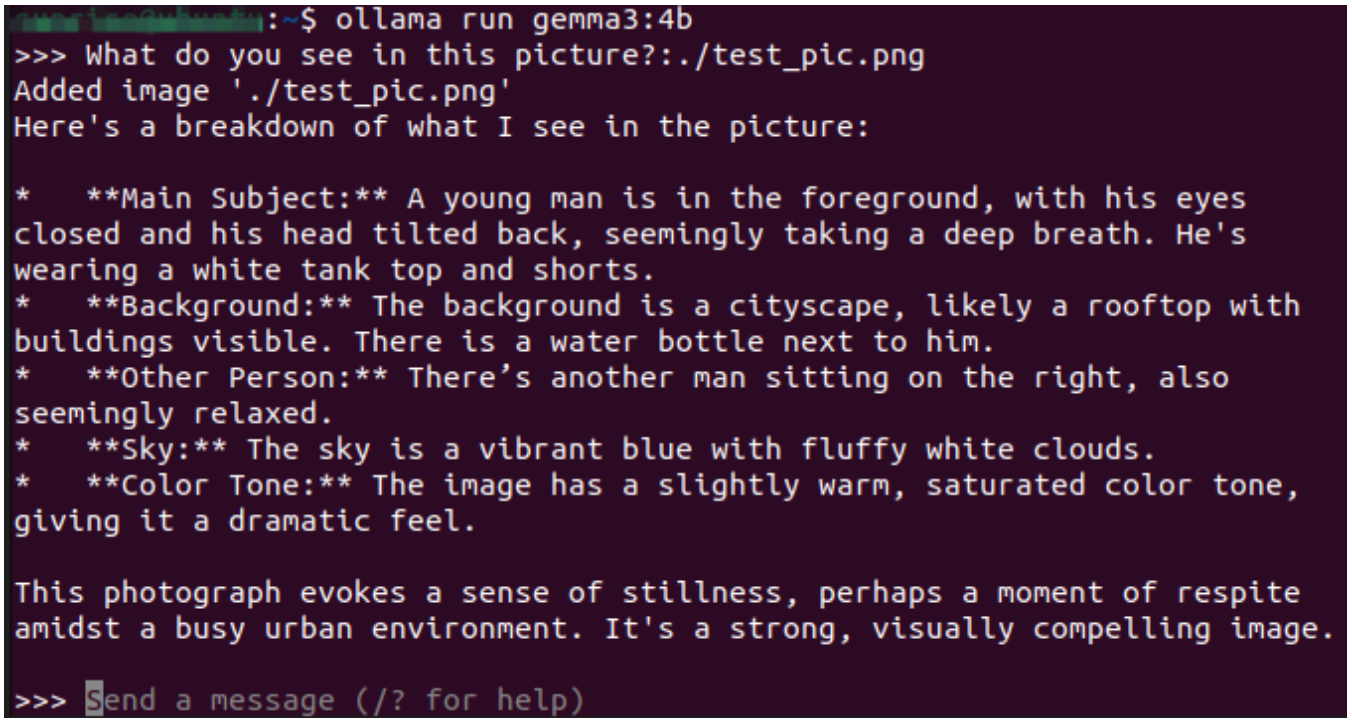
**1. Choosing Your Language:**

* **Beginner-Friendly Options:**
    * **Python:**  Highly recommended for beginners. It has a clean
syntax, huge community support, and is used in many fields (web
development, data science, scripting).
    * **JavaScript:**  Essential for web development (front-end and
increasingly back-end with Node.js).
    * **Scratch:** (Visual, block-based) A fantastic starting point for
kids and absolute beginners to understand programming concepts without
worrying about syntax.
* **Consider Your Goals:** What do you want to *do* with programming?
    * **Web Development:** JavaScript, Python (with frameworks like Django
or Flask), Ruby
    * **Data Science/Machine Learning:** Python, R
    * **Game Development:** C#, C++, Python (with Pygame)
    * **Mobile App Development:** Swift (iOS), Kotlin/Java (Android)
```

3.3 Vision function



```
What do you see in this picture? :./test_pic.png
# Using ": + the absolute path of the image" in the conversation lets the model use its
vision function and parse the information in the image
```



3.4 End the conversation

Use the `Ctrl+d` shortcut or `/bye` to end the conversation!

References

> **Ollama**

Official website: https://ollama.com/

GitHub: https://github.com/ollama/ollama

> Gemma3

Corresponding Ollama model: https://ollama.com/library/gemma3